



CLINICAL GENETICS

Problem solving



Case 1

- A 7-year-old boy presents with short stature, congenital heart disease (pulmonary valve stenosis), hypertelorism, and webbed neck.

- *What is the most likely diagnosis?*
- *What is the mode of inheritance?*
- *What genetic test confirms the diagnosis?*



- *Diagnosis: Noonan syndrome*
- *Inheritance: Autosomal dominant*
- *Diagnostic test: PTPN11 gene sequencing or multigene panel*

Case 2

- A tall boy with long limbs, arachnodactyly, pectus deformity, and aortic root dilation is evaluated in clinic.
 - *What is the most likely diagnosis?*
 - *Which gene is commonly mutated?*
 - *What is the inheritance pattern?*



- *Diagnosis: Marfan syndrome*
- *Gene: FBN1*
- *Inheritance: Autosomal dominant*

Case 3

- A newborn has rhizomelic limb shortening, frontal bossing, and midface hypoplasia.

- *What is the most likely diagnosis?*
- *Which gene is responsible?*
- *What is the inheritance pattern?*



- *Diagnosis: Achondroplasia*
- *Mutation: FGFR3*
- *Inheritance: Autosomal dominant*

Case 4

An infant presents with hypoglycemia, lactic acidosis, hepatomegaly, and doll-like facies.

- *What is the diagnosis?*
- *What is the treatment modality ?*

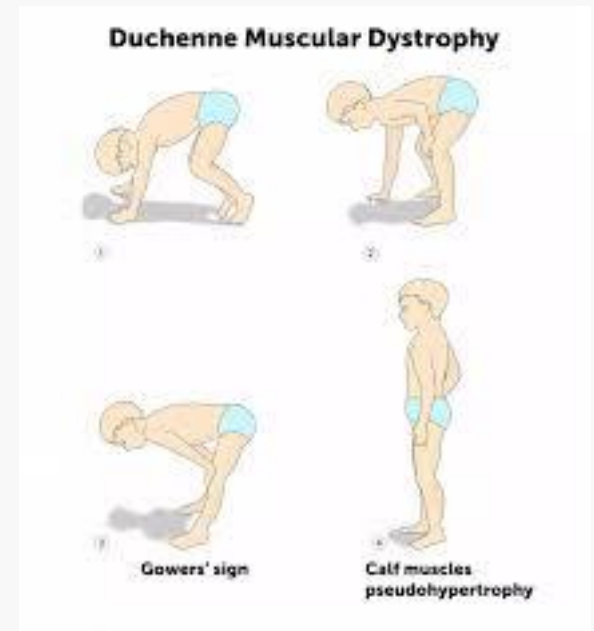


- *Diagnosis: Glycogen storage disease type I*
- *Frequent feeding avoiding hypoglycemia, night time feeding , Corne starch , liver transplant .*

Case 5

- A 4-year-old boy has progressive muscle weakness, calf pseudohypertrophy, and difficulty climbing stairs.

- *What is the diagnosis?*
- *What is the inheritance pattern?*
- *What laboratory test is typically elevated?*



- *Diagnosis: Duchenne muscular dystrophy*
- *Inheritance: X-linked recessive*
- *Lab test: Elevated creatine kinase (CK)*

Case 6

- A school-age boy has intellectual disability, large ears, elongated face, and behavioral difficulties.
 - *What is the diagnosis?*
 - *What genetic abnormality causes this syndrome?*
 - *What is the inheritance pattern?*



- *Diagnosis: Fragile X syndrome*
- *Abnormality: CGG trinucleotide repeat expansion in FMR1*
- *Inheritance: X-linked dominant*

Case 7

- A newborn has hypotonia, upslanting palpebral fissures, single palmar crease, and congenital heart disease.
 - *What is the diagnosis?*
 - *What chromosomal abnormality is present?*
 - *What test confirms the diagnosis?*



- *Diagnosis: Down syndrome*
- *Abnormality: Trisomy 21*
- *Diagnostic test: Karyotype analysis*

Case 8

- A newborn presents with cleft lip/palate, microphthalmia, polydactyly, and severe developmental abnormalities.
 - *What is the diagnosis?*
 - *What chromosomal abnormality causes it?*
 - *What test confirms the diagnosis?*



- *Diagnosis: Patau syndrome*
- *Abnormality: Trisomy 13*
- *Diagnostic test: Karyotype analysis*

Case 9

- A teenage girl has short stature, primary amenorrhea, webbed neck, and lymphedema.
 - *What is the diagnosis?*
 - *What is the classic karyotype?*
 - *What cardiovascular abnormality is commonly associated?*



- *Diagnosis: Turner syndrome*
- *Karyotype: 45,X*
- *Association: Coarctation of the aorta*

Case 10

- An adolescent boy has tall stature, gynecomastia, infertility, and small testes.
 - *What is the diagnosis?*
 - *What is the karyotype?*
 - *What hormone profile is expected?*



- *Diagnosis: Klinefelter syndrome*
- *Karyotype: 47,XXY*
- *Hormones: Low testosterone with elevated FSH/LH*

Case 11

- An infant has a high-pitched cat-like cry, microcephaly, and developmental delay.
 - *What is the diagnosis?*
 - *What chromosomal abnormality is present?*
 - *What test confirms the diagnosis?*



- *Diagnosis: Cri-du-chat syndrome*
- *Abnormality: Deletion of chromosome 5p*
- *Diagnostic test: Karyotype ,FISH, chromosomal microarray*

Case 12

- A child presents with neonatal hypotonia followed later by hyperphagia, obesity, and intellectual disability.

- *What is the diagnosis?*
- *What is the genetic mechanism?*
- *What molecular test is used for diagnosis?*



- *Diagnosis: Prader-Willi syndrome*
- *Mechanism: Loss of paternal 15q11-q13 expression*
- *Diagnostic test: DNA methylation analysis , FISH , microarray*

Case 13

- A child has severe developmental delay, seizures, ataxia, frequent laughter, and minimal speech.
 - *What is the diagnosis?*
 - *What is the genetic mechanism?*
 - *What molecular test is used for diagnosis?*

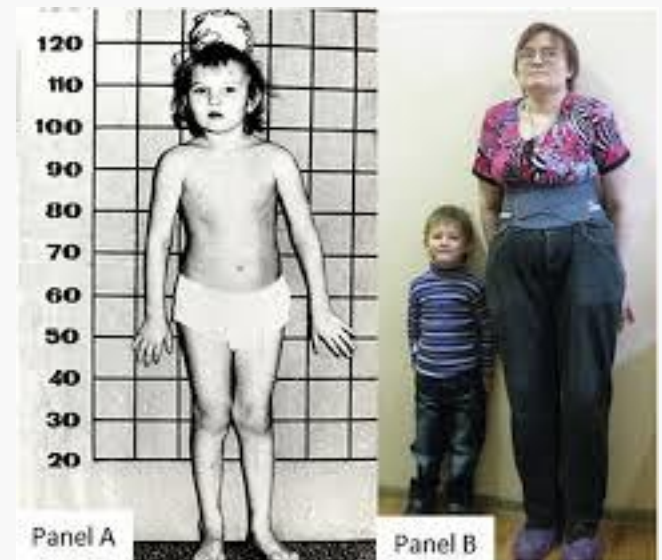


- *Diagnosis: Angelman syndrome*
- *Mechanism: Loss of maternal UBE3A expression in 15q11-q13*
- *Diagnostic test: DNA methylation analysis and UBE3A testing*

Case 14

A child presents with developmental delay, marfanoid habitus, lens dislocation downward, and thromboembolic events.

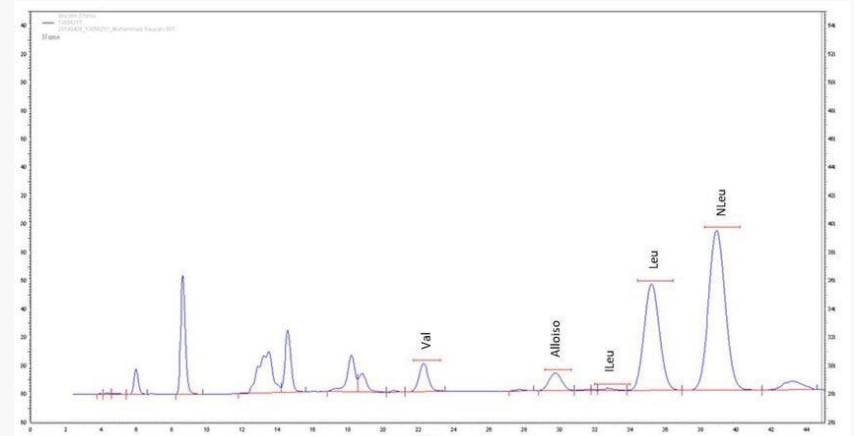
- *What is the diagnosis?*
- *What is the mode of inheritance ?*
- *What laboratory finding is expected?*



- *Diagnosis: Homocystinuria*
- *AR*
- *Lab finding: Elevated homocysteine and methionine levels*

Case 15

- A neonate develops poor feeding, lethargy, and urine with a sweet maple syrup odor.
 - *What is the diagnosis?*
 - *What amino acids accumulate?*



- *Diagnosis: Maple syrup urine disease*
- *Accumulated amino acids: Leucine, isoleucine, alloisoleucine and valine*

Case 16

A newborn develops vomiting, lethargy, respiratory alkalosis, and hyperammonemia shortly after birth.

- *What type of disorder is suspected?*
- *What laboratory abnormality is characteristic?*
- *What emergency treatment is required?*

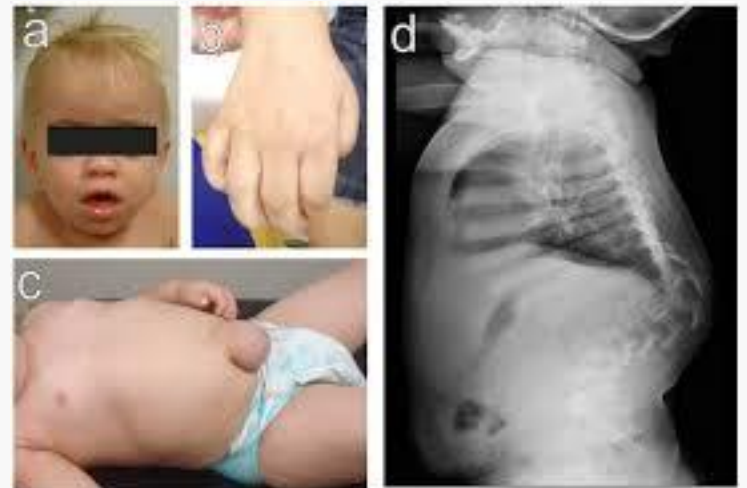
- *Diagnosis: Urea cycle disorder*
- *Characteristic abnormality: elevated Glutamine, elevated or low citrulline ,elevated or low urine orotic acid.*
- *Treatment: Protein restriction and ammonia-lowering therapy*

Case 17



A child presents with coarse facial features, hepatosplenomegaly, developmental delay, and skeletal abnormalities.

- *What is the diagnosis category?*
- *What substance accumulates in these disorders?*
- *What diagnostic test is useful?*



- *Diagnosis: Mucopolysaccharidosis*
- *Accumulated substance: Glycosaminoglycans (GAGs)*
- *Diagnostic test: Urine GAG analysis and enzyme assay*

Case 18

A neonate develops jaundice, vomiting, hepatomegaly, and cataracts after milk feeding.

- *What is the diagnosis?*
- *Which enzyme deficiency is responsible?*
- *What dietary treatment is required?*



- *Diagnosis: Classic galactosemia*
- *Enzyme deficiency: Galactose-1-phosphate uridylyltransferase deficiency*
- *Treatment: Elimination of lactose and galactose from diet*